Foundations of Data Engineering and Analytics

Data Analysis Project: Concept

Lucas Alois Griesser, Luca Jenewein, Nikola Adžić · Winter Semester 2026/27

Forecasting the Net Battery Energy of a Formula Student Electric Car

Research question

How accurately can the net battery energy of the CT7 over the next 5 seconds of driving be predicted from the driver inputs and vehicle state of the preceding 5 seconds, and do standard machine-learning models outperform a persistence baseline and a physics-based baseline on test days that were not used for training?

The CT7 is the 2025 electric race car of the Formula Student team CTM. The *net battery energy* of a window is $E = \sum_k P_k \, \Delta t$ with $\Delta t = 0.1\,\mathrm{s}$, where P is the power at the battery output reported by the car's ISA current and voltage sensor (it agrees closely with $U \cdot I$). E is positive when the battery is discharged and negative when regenerative braking feeds back more energy than the car uses in that window.

Hypothesis test: H_0: on the held-out test days, the median difference between the day-level mean absolute error of the tuned machine-learning model (model and hyperparameters chosen by grouped cross-validation on the training days only) and that of the physics-based baseline is zero. H_1: the median difference is not zero. Test: two-sided Wilcoxon signed-rank test on the paired day-level errors, $\alpha = 0.05$.

Why the question is interesting and challenging

In the endurance event the battery capacity limits how fast the car can be driven over the whole distance, so the team needs to know how much energy short sections of driving cost. A short-term forecast is a building block for energy management (power limits, regeneration settings) and for feedback to the drivers. The question is not trivial: the next 5 seconds depend on what the driver does next, on the track and on the car's settings, none of which is known in advance. The data are real logs with clock jumps, firmware changes and corrupted values, and samples within a drive are not independent, so an honest evaluation must keep whole test days out of training.

Related work

De Cauwer et al. [1] predict the energy consumption of an electric passenger car (a Nissan Leaf) from real-world driving data with multiple linear regression on the terms of the vehicle dynamics equation (rolling resistance, potential energy, aerodynamic losses, kinetic energy). Huang et al. [2] combine a physics-based simulation model with random forests and boosting that predict the physical model's residuals; on real trips this decreased the average prediction error from 40 % for the pure physics model to 10 %, evaluated with leave-one-trip-out cross-validation. Roberts et al. [3] show that random cross-validation underestimates the prediction error when data are dependent and recommend block cross-validation. This project transfers these ideas to a race car: a short horizon of 5 seconds, driver inputs as features, a physics-based baseline, and validation by whole test days.

Data

The data are CAN bus logs that a Raspberry Pi in the car recorded with `candump` on four buses during test drives between May and August 2025 (about 31 GiB, stored by CTM). CTM allowed their use for this project and the submission of a raw excerpt with the message definitions. A one-time scan of the 342 non-empty files named 26 May 2025 or later (27.1 GiB; earlier logs contain no driving with a battery power measurement) kept the frames of 10 messages, at most one per message and 100 ms, while a wheel motor turned faster than 100 1/min, plus 30 s before and after. Both notebooks run from this submitted excerpt (5.8 million frames, 36.5 MiB). The extraction notebook decodes 12 signals from it: the four wheel motor speeds, accelerator and brake pedal positions, regeneration request, steering angle, battery current, voltage and power, and the maximum battery cell temperature. The pre-processing notebook turns them into pairs of consecutive 5-second windows: features from window w, the net battery energy of window $w{+}1$ as target. The resulting dataset contains 4,401 window pairs from 315 stints, about 6.8 hours of driving under power on 11 test days; the target ranges from -53 to 226 kJ. Known limitations of the data: no ambient temperature, no position or lap information, and the steering angle only from 21 June 2025.

Methods, resources and evaluation

- **Pre-processing:** timestamps and clock jumps, invalid values, alignment on a 10 Hz grid, consistency of P and $U \cdot I$, stint segmentation, window features.
- **Exploratory analysis:** distributions of target and features, relations between energy, motor speed and pedal use, correlations, and differences between test days.
- **Baselines:** mean of the training days; persistence (the energy of the next window equals that of the current one); a physics-based linear regression on terms of the vehicle dynamics equation, computed from the motor speed n as a speed proxy ($n \cdot \Delta n$, n, n^3).
- **Models:** random forest and histogram-based gradient boosting from scikit-learn, compared with two feature sets (driver inputs and speed; additionally battery state and the current window energy) and a small grid of hyperparameters, tuned with grouped cross-validation over the training days.
- **Evaluation:** leave-one-day-out cross-validation; mean absolute error, root mean squared error and R^2, also per day and per speed range; the hypothesis test above.
- **Resources:** Python, pandas, scikit-learn, SciPy and Jupyter on a laptop.

Risks and limitations

Only 11 test days are available, and they contribute very different amounts of data, so the day-level hypothesis test has little statistical power and results may vary strongly between days. The days differ in track, driver, weather and car settings in ways that are not recorded, which can make the models fail on a new day. All data come from test drives, not from competition events. The speed is approximated from the motor speed, which is affected by wheel slip. The results describe one car and cannot simply be transferred to other vehicles.

References

[1] C. De Cauwer, J. Van Mierlo, T. Coosemans. Energy consumption prediction for electric vehicles based on real-world data. *Energies* 8(8):8573–8593, 2015. `https://doi.org/10.3390/en8088573`

[2] Y.-W. Huang, C. Prehofer, W. Lindskog, R. Puts, P. Mosca, G. Kauermann. Predictive energy management for battery electric vehicles with hybrid models. arXiv:2405.10984, 2024.

[3] D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40:913–929, 2017. `https://doi.org/10.1111/ecog.02881`